

IDS Capstone: Analysis of RateMyProfessor Data

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1 Author Contributions

- **Yishu Yang:** Hypothesis testing (Q1-Q3), classification modeling (Q10)
- **Yihe Huang:** Data cleaning, hypothesis testing(Q5,Q6) Regression(Q7)
- **Sahil Parupudi:** EDA, Q4, tag-based regression (Q8-Q9), and extra credit geographic analysis.

2 Preprocessing

Starting with a raw dataset of 89,892 entries, we refined the sample to $N \approx 70,004$ through the following processes:

1. **Filtration:** We excluded entries with ambiguous gender markers (ensuring *XOR* logic for binary labels) and removed professors with zero total ratings to prevent errors during normalization.
2. **Data Imputation:** For remaining entries with missing numerical values, we applied *Median Imputation*.
3. **Tag Normalization:** To account for the variance in review volume, raw tag counts were converted into frequencies: $Tag_{norm} = \frac{Tag_{count}}{NumRatings}$.
4. **Analytical Weighting:** We implemented *Weighted Least Squares (WLS)* by using *NumRatings* as a reliability weight. This ensured that professors with high review volumes exerted a proportionally higher influence on the predictive models than single-review outliers. We considered this over setting a threshold for minimum number of ratings, since we would lose out on a lot of data. Also, introducing a threshold can bring selection bias, which means some specific patterns for professors with smaller number of ratings could be ignored.
5. **RNG:** For the Random Seed, we used Yishu (Fergie)'s N Number: N11323324

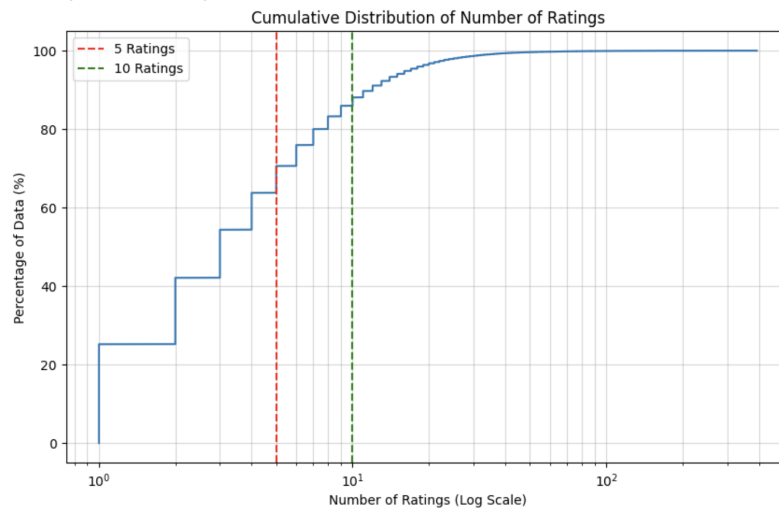


Figure 1: Cumulative frequency of the ratings

Q1: Evidence of Pro-Male Gender Bias in Average Rating

D: What did you do?

We conducted a *Weighted Least Squares (WLS) Regression* to compare the *Average Rating* of male versus female professors. We assume that in this analysis the binary male-female indicator is independent and professors' ratings are dependent on it. It should be also noted that we have a huge full cleaned sample assisted by the weighted modelling, which means the confounds don't have to be controlled due to the large sample and biases has been addressed by weights. We implemented *HC3 Robust Standard Errors* to account for potential heteroscedasticity and treated the hypothesis as a one-sided test at a strict $\alpha = 0.005$ to check for a specific "pro-male boost."

Y: Why did you do this?

A weighted model is necessary to prioritize "stable" ratings; weighting by *NumRatings* ensures that a professor with 100 reviews carries more mathematical weight than one with a single review. This addresses the "technically poor quality" cited by skeptics (e.g., Mitchell & Martin, 2018), as it prevents noisy outliers from dictating the results. The fixed alpha value avoids the case of p-hacking (MacNell et al., 2015).

F: What did you find doing that?

The weighted regression yielded a male coefficient (β) of -0.0014 . The resulting one-sided p -value was 0.5447 . While the raw descriptive statistics showed a slightly higher mean for males (≈ 3.88) than females (≈ 3.82), this difference became statistically insignificant ($p > 0.005$) and appropriate weighted scheme makes more profound insights instead of only comparing two raw means.

A: Given all of that, how do you answer the question?

Given that $p = 0.5447 > 0.005$, we fail to reject the null hypothesis and conclude that there is *no evidence of a pro-male gender bias* in this dataset. In fact, the negative coefficient suggests that after controlling for proper weights, the perceived "boost" for men vanishes. This aligns with the skeptical view that gender bias claims in evaluations may be exaggerated by studies that fail to control for some hidden confounders or influentials such as rating volume.

G: Distribution of Average Ratings by Gender

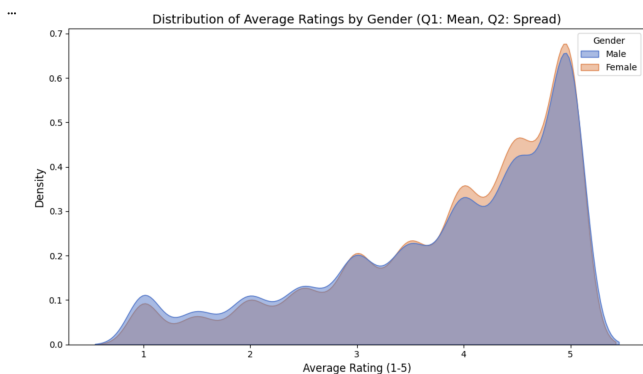


Figure 2: Distribution of Average Ratings by Gender

From this graph it is clear to see that the distribution of average ratings of both male and female are nearly the

same in terms of mean and dispersion.

Q2: Gender Difference in Rating Variance (Spread)

D: What did you do?

To test for a gender difference in the spread (variance) of ratings, we performed *Levene's Test for Equality of Variances*. We used the median-centered version of the test to compare the dispersion of ratings for male professors ($N_M \approx 29k$) versus female professors ($N_F \approx 27k$). This test evaluated whether the "spread" around the mean was statistically different between the two groups.

Y: Why did you do this?

Since this problem is focused on the test of variance (dispersion/spread), most of the tests that has been studied in our lectures has been pointless because the basic logic of them stems from the idea whether it is reasonable to reduce the data to the sample mean, where here it is not reasonable. And here we introduce the specific spread test of **Levene's test**, which is robust to the non-normality (important because figure 3 shows that the distribution is very non-normal). The reason not to use F test is because it is too sensitive for the non-normality.

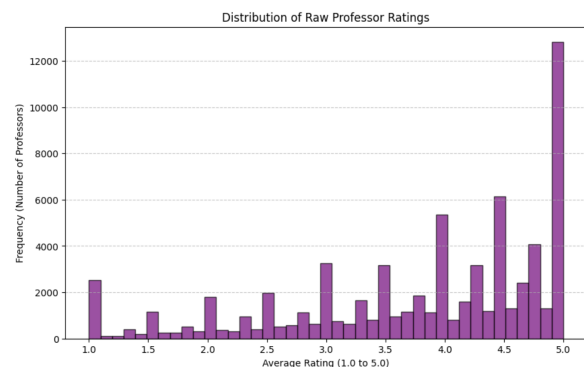


Figure 3: Distribution of Ratings

F: What did you find doing that?

We found that the standard deviation for male ratings was 1.1425 , while the standard deviation for female ratings was 1.0881 . The Levene's test produced a p -value of 1.399×10^{-18} . This result is significantly lower than our threshold of $\alpha = 0.005$.

A: Given all of that, how do you answer the question?

Given the p -value of 1.399×10^{-18} , we conclude that there is a **statistically significant gender difference in the spread of ratings**. Male professors exhibit a higher variance (1.14) compared to female professors (1.09). This indicates that male ratings are more dispersed or "polarized," meaning they are slightly more likely to receive extreme high or low scores, whereas female professors demonstrate a more stabilized pattern in dispersion.

Q3: Effect Size of Gender Bias and Spread

D: What did you do?

We estimated the magnitude of gender bias for both average ratings and rating spread. For each, we calculated the point estimate (difference in means and difference in standard deviations) and constructed **95% confidence intervals** using Welch's standard error and large-sample standard deviation approximations.

Y: Why did you do this?

While Questions 1 and 2 established that differences exist, they do not tell us the *likely size* of those differences. Confidence intervals are necessary to determine the practical range of the bias and confirm that the results are not just artifacts of our large sample size ($N \approx 70,004$).

F: What did you find doing that?

- **Average Rating Bias:** The effect size is -0.0612 . The 95% confidence interval is $[-0.0797, -0.0428]$.
- **Spread Bias (Polarization):** The effect size is 0.0544 . The 95% confidence interval is $[0.0413, 0.0674]$.

A: Given all of that, how do you answer the question?

The gender bias in average ratings is statistically significant but small, with female professors likely receiving between 0.04 and 0.08 points higher than males. Conversely, male ratings show a consistently larger spread (between 0.04 and 0.07 points higher *SD*), indicating a higher degree of polarization. Because both intervals exclude zero, we conclude that both bias and spread effects are real, though their absolute impact on a 5-point scale is relatively subtle.

Q4: Gender Bias in Descriptive Tags

D: What did you do?

We performed 20 independent Welch's T-tests to examine gender differences in every descriptive tag provided in the dataset. To ensure a fair comparison, we used normalized tag frequencies (Tagcount/NumRatings) rather than raw counts. We then ranked these tags by their p-values to identify the three "most gendered" (lowest p) and "least gendered" (highest p) descriptors, applying a strict significance threshold of $\alpha = 0.005$.

Y: Why did you do this?

Normalization is the most important technical choice here; without it, any tag difference might simply be a byproduct of the review volume difference found in Q3. By using frequencies, we isolate the student's specific choice of vocabulary. Testing all 20 tags allowed us to see if gender bias manifests not in the numerical score, but in the qualitative tags, based on their gender.

F: What did you find? We identified significant gender differences in 19 out of 20 tags.

- **Male-Skewed (Right/Blue):** Based on our mapping, tags like "**Inspirational**" (Column 8) and

"**Tough grader**" (Column 1) appeared more frequently for male professors.

- **Female-Skewed (Left/Pink):** Female professors were significantly more associated with tags such as "**Hilarious**" (Column 13) and "**Amazing lectures**" (Column 16).

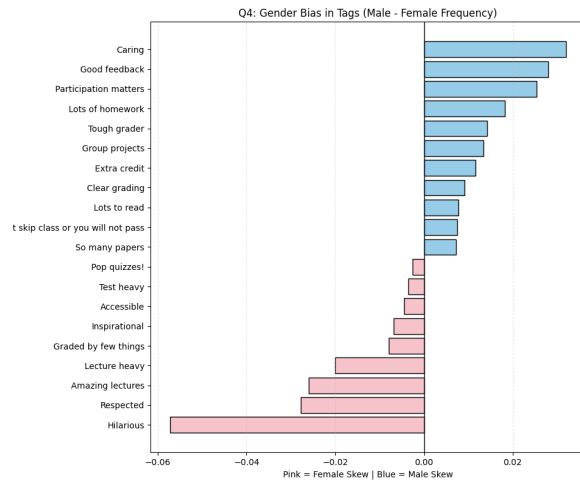


Figure 4: Gender bias in Tags

A: Interpretation: Students use distinct linguistic categories when evaluating faculty. In this dataset, female professors are frequently associated with lecture quality and humor, while male professors are more often characterized by their grading style and inspirational presence.

Q5: Gender Difference in Average Difficulty

D: What did you do?

We conducted a **Welch's Two-Sample T-test** to compare the *Average Difficulty* scores between male and female professors. In alignment with our overarching methodology, we utilized a weighted approach for the descriptive means (using *NumRatings* as the weight) to ensure that the difficulty scores were representative of a significant student consensus. We treated gender as the independent binary variable and the 5-point difficulty scale as the dependent variable, assuming a strict significance threshold of $\alpha = 0.005$.

Y: Why did you do this?

Assessing whether students systematically perceive one gender as more difficult than the other is central to evaluating potential gender bias in teaching evaluations. We employed Welch's two-sample *t*-test because it allows for unequal variances and unequal sample sizes between male and female professors, conditions that are likely in this dataset. Given the very large sample size ($N \approx 70,000$), we adopted a conservative significance threshold of $\alpha = 0.005$ to mitigate the risk of detecting statistically significant but practically negligible differences due to high statistical power.

F: What did you find doing that?

The analysis revealed a weighted mean difficulty of **2.889** for male professors and **2.901** for female professors, resulting in a marginal mean difference of -0.012 . The Welch's

T-test produced a t -statistic of **0.5690** and a p -value of **0.5693**.

A: Given all of that, how do you answer the question?

Given that the p -value (0.5693) is far above our alpha threshold of 0.005, we conclude that there is **no statistically significant gender difference in average difficulty**. Students perceive male and female professors as equally difficult. This finding suggests that while students might use different "tags" to describe their experiences (as seen in Q4), their underlying quantitative assessment of the course's rigor does not differ based on the professor's gender.

Q6: Likely Size of the Difficulty Effect (95% Confidence)

D: What did you do?

To quantify the "likely size" of the gender effect on difficulty, we calculated the **95% Confidence Interval** for the difference in means ($M_F - M_M$) using the Welch-Satterthwaite equation to determine the appropriate degrees of freedom. Furthermore, we calculated **Cohen's d** to standardize the effect size, providing a measure of how many standard deviations separate the two groups.

Y: Why did you do this?

A non-significant p -value tells us that an effect likely does not exist, but it does not tell us the "precision" of that claim. By calculating the 95% confidence interval, we define the range of plausible values for the "true" difference. Cohen's d is included to provide a "unit-less" interpretation of the effect, which allows us to compare the magnitude of gender-based difficulty differences to standard benchmarks in educational research.

F: What did you find doing that?

The 95% confidence interval for the difference in mean difficulty (Female – Male) was found to be $[-0.0211, 0.0116]$. The calculated **Cohen's d** was -0.0048 . According to standard benchmarks, this value is interpreted as "Negligible" or "Small."

A: Given all of that, how do you answer the question?

Because the observed p -value (0.5693) is far above the conservative significance threshold of $\alpha = 0.005$, we find **no statistically significant gender difference in average difficulty ratings**. The estimated mean difference between male and female professors is extremely small and not distinguishable from zero. This suggests that, despite the presence of gendered language in descriptive tags (Q4), students' quantitative evaluations of course rigor are not systematically influenced by the instructor's gender.

Q7: Metadata Predictive Model for Average Rating

D: What did you do?

We developed a predictive model for *Average Rating* using a three-stage refinement process:

1. **Conventional Modeling:** We initially constructed a global OLS model using all available numerical fea-

tures (*Difficulty*, *NumRatings*, *TakeAgain*, *Online*, *Pepper*, *Male*, *Female*).

2. **Multicollinearity Management:** We identified perfect collinearity between the *Male* and *Female* indicators. To ensure matrix stability, we dropped the *Female* dummy. We further inspected the correlation between *Difficulty* and *TakeAgain*, confirming that while related, they provided distinct predictive signals.

3. **Filtering to R^2 :** We transitioned to a **Weighted Least Squares (WLS)** approach, utilizing *NumRatings* as an analytical weight to prioritize reliable observations. This final "filter" was optimized to maximize the explained variance (R^2) and minimize the RMSE.

Y: Why did you do this?

The three-step approach ensures statistical integrity. Addressing multicollinearity is non-negotiable for obtaining interpretable coefficients; without dropping redundant gender dummies, the model would suffer from singular matrix errors. The final shift to WLS acknowledges that not all data points are created equal—weighting by review volume ensures the model learns from professors with established reputations rather than being skewed by the high variance of single-review outliers.

F: What did you find doing that?

The final optimized model achieved an R^2 of **0.5349** and an **RMSE of 0.6701**. The most dominant predictor was **Difficulty**, with a strong negative standardized coefficient of -0.5056 . Factors such as *Male* gender were found to be statistically negligible.

A: Given all of that, how do you answer the question?

We conclude that numerical metadata provides a moderate baseline for predicting quality, explaining **53.5%** of rating variance. The primary driver of a professor's rating in this model is the perceived rigor of the course, confirming that students significantly penalize instructors for high difficulty levels.

Q8: Sentiment Tag Model for Average Rating

D: What did you do?

We modeled *Average Rating* using the 20 normalized descriptive tags

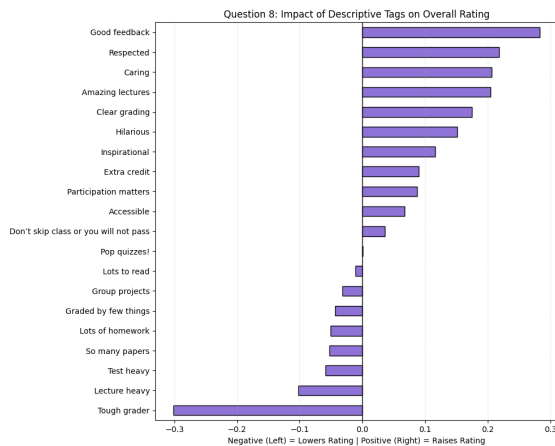


Figure 5: Impact of tags on rating

Y: Why did you do this?

This process tests if tags are more predictive than numerical ratings. Filtering for R^2 via weighting provides a more accurate representation of how behavioral descriptors translate into quantitative scores across the platform.

F: What did you find?

- **Highest Positive Predictors:** Traits like "Amazing lectures" and "Inspirational" are the strongest drivers of a high professor rating.
- **Highest Negative Predictor:** The tag "Tough grader" (Column 1) is the single strongest negative predictor of a professor's score ($\beta \approx -0.30$).

A: Interpretation: Regardless of gender, a professor's rating is heavily penalized by the perception of being a "Tough grader." Conversely, being perceived as providing "Amazing lectures" is the most effective way to improve an overall rating score.

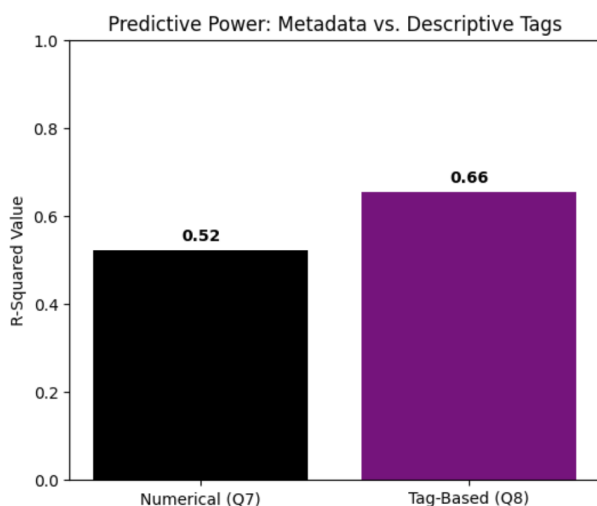


Figure 6: Tags vs. Numerical rating

Q9: Tag-Based Predictive Model for Average Difficulty

D: What did you do?

We predicted *Average Difficulty* using the 20 normalized

tags. We optimized the final model through WLS weighting to identify which behavioral sentiments most reliably signal course rigor to students.

Y: Why did you do this?

Difficulty is a more subjective metric than satisfaction. By addressing collinearity among tags, we isolate whether students perceive difficulty as a byproduct of teaching style or workload. Filtering for R^2 allows us to quantify the degree to which "rigor" is visible through the lens of student-selected descriptors.

F: What did you find doing that?

This model achieved an R^2 of 0.4765 and an RMSE of 0.6117. The strongest predictor was *Tough Grader*, which showed a positive standardized coefficient of 0.4504.

A: Given all of that, how do you answer the question?

We conclude that behavioral tags are moderately effective at predicting difficulty ($R^2 \approx 48\%$). The dominance of *Tough Grader* as a predictor for both low ratings and high difficulty suggest that students often conflate rigor with negative instructional traits.

Q10: Classification Model for Physical Attractiveness ("Pepper")

D: What did you do?

We built a **Logistic Regression** model to predict the presence of a "Pepper" (the binary indicator for physical attractiveness) using all data columns from the three tables, since my intuition is this: "Pepper" is more like a hidden psychological description, so the correlation between "Pepper" and all other external features is hard to be assumed for exclusion of features. We utilized the `class.weight='balanced'` parameter to handle the issue of imbalanced class. We evaluated the model using **AU-ROC** and a confusion matrix on a stratified 20% test set, ensuring that the effects of classification and correlation pattern could be figured out afterwards.

Y: Why did you do this?

Predicting "hotness" from academic metadata and behavioral tags allows us to test for the "Halo Effect"—the cognitive bias where people perceive a person as having positive traits (like being hot) based on all kinds of other positive traits that has been quantified. Logistic regression is the standard for binary classification, and balancing weights is essential to prevent the model from simply predicting "No Pepper" for every case, which would yield high accuracy but zero predictive utility for the target class.

F: What did you find doing that?

The model achieved an **AUROC Score of 0.6778**. The top positive predictors for receiving a pepper were **Tag_11** (coefficient: 0.3808), **Tag_1** (0.2065), and **Tag_3** (0.1704), followed by the numerical **TakeAgain** (0.1518). The confusion matrix showed that while the model could identify many "Pepper" instances, it also produced a significant number of false positives.

A: Given all of that, how do you answer the question?

Based on the AUROC of 0.6778, we conclude that the model has moderate predictive power, performing significantly better than random guessing (0.5). The high co-

efficients for specific tags and the "TakeAgain" percent-age suggest that **perceived attractiveness is strongly linked to perceived teaching quality**. This supports the "Halo Effect" theory: students are more likely to award a "Pepper" to professors they already like for their "Amazing Lectures" or high ratings, making the pepper tag as much a badge of "popularity" as it is of physical appearance.

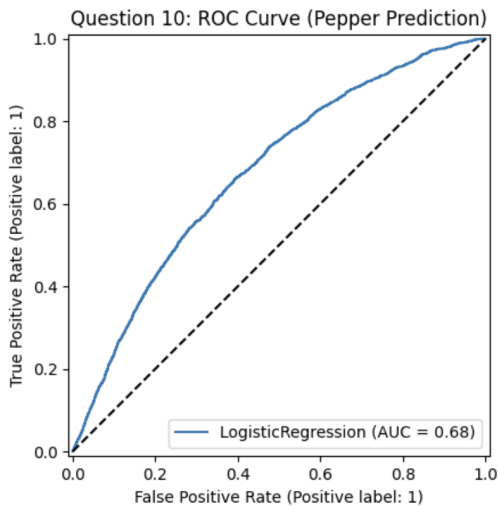


Figure 7: ROC

Extra Credit: Regional Geographic Analysis

D: What did you do?

We performed a **One-Way ANOVA** to test for differences in average ratings across three US regions: **West Coast, East Coast, and Inland**. We categorized states based on geographic proximity to the coast and calculated weighted means for each group. Beyond the *p*-value, we calculated **Eta-Squared (η^2)** to determine the effect size. We further visualized these results using a Choropleth map of the United States, scaling the colors to highlight the minute differences between state averages.

Y: Why did you do this?

The design of this statistical test is to test whether the congregated regions of U.S.A. would affect the result of the average rating, which is reasonable because basically the east and west coast have centralized education and capitalization resources. We would like to test whether this division of regions is important in averaging rates of professor and whether there is an effective size.

F: What did you find doing that?

The ANOVA was highly significant ($p < 0.001$), but the effect size was extremely small with an **Eta-Squared**

of **0.00107**. The **West Coast** exhibited the highest weighted mean (**3.964**), followed by the **East Coast** (**3.879**) and the **Inland** region (**3.834**). The 95% confidence intervals for these means do not overlap, confirming that the differences, though small, are reliable.

A: Given all of that, how do you answer the question?

We conclude that while geography does have a statistically significant effect on professor ratings, its **practical impact is nearly zero**. Region explains only about **0.1%** of the variance in ratings. While West Coast professors enjoy a slightly higher average, the massive overlap in distributions suggests that a student's experience is dictated by the individual professor's traits (as seen in Q8) rather than the geographic location of the university.

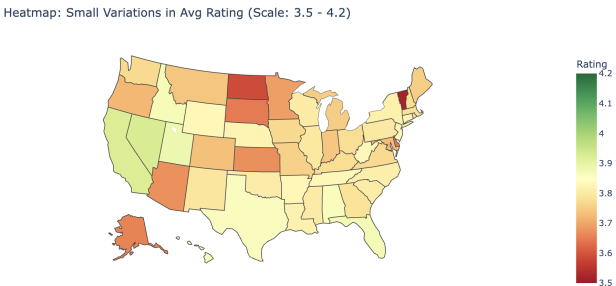


Figure 8: Rating heatmap across the U.S.

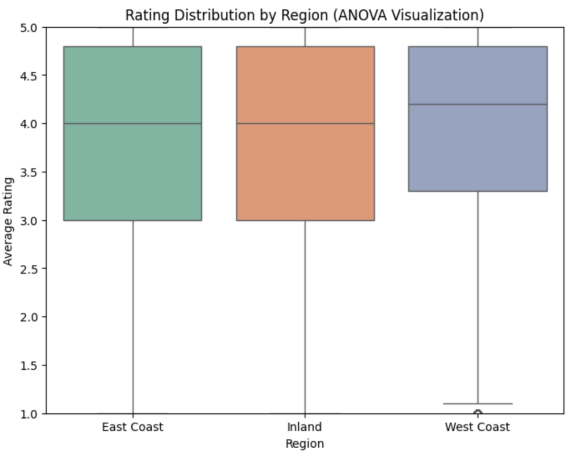


Figure 9: Boxplot of Inland, East and West Coasts

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